

Leveraging Machine Learning to Analyze Eviction Patterns in the DMV

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ABSTRACT

In 2018, 3.6 million eviction cases were filed nationwide. More than double that number of individuals are threatened by eviction every year, many of them children. An eviction is often a traumatic event for a family, causing long term financial, health, and employment impacts. Even so, data on evictions is limited and unreliable due to a lack of standardization across states, making policy interventions difficult to support empirically. In this paper, I examine evictions within the Washington, D.C. metropolitan area and attempt to apply machine learning methods to explain and predict eviction filing rates. This pursuit may contribute to literature exploring populations at greatest risk of eviction, as well as providing a method for identifying areas for potential targeted policy intervention. Ultimately, I employ several classification models and identify significant predictors of eviction within a simple logistic regression pipeline.

Keywords: Eviction, Machine Learning, Classification

INTRODUCTION

Due to rising costs and stagnating wages, housing is one of the largest costs faced by American families today. This financial burden contributes to growing housing insecurity and increasing eviction rates across the nation (Desmond, 2015). The ramifications of an eviction are long-lasting and far-reaching: an eviction record makes it harder to find future housing, increases health risks, damages children's education, often causes unemployment, and may result in temporary or long-term homelessness. Despite these devastating realities for millions of families each year, data on evictions unreliable and difficult to access, if it is collected at all. Additionally, eviction is closely related to other facets of poverty, including race, immigration status, earnings potential, education, and other socioeconomic indicators. As such, eviction is not only an issue of economics, but of social justice; it is both a cause and effect of poverty.

Machine learning has been employed to better understand housing, eviction, and related issues in several instances, useful for its capacity to handle highly correlated socioeconomic indicators, though also limited by poor data on evictions. As eviction is highly influenced by local and state laws, studies are localized and often non-generalizable, necessitating individual attention to specific geographies. With these points in mind, I utilize data from the Princeton Eviction Lab and the American Community Survey to model eviction rates in the Washington D.C. metropolitan area. Specifically, I examine what communities are at a highest risk of eviction and what socioeconomic indicators are most influential in this determination. While these questions have been examined thoroughly across sociological and economic perspectives, machine learning tools may identify key features previously overlooked and act as a critical forecasting tool, informing not only responses to eviction, but also prevention efforts.

BACKGROUND

How Does an Eviction Happen and What are the Consequences?

A landlord cannot legally remove their tenants at a whim; they must follow the legal proceedings for an eviction according to their local and state laws. This usually begins with a documented notice of violation—such as non-payment of rent—and if the tenant continues to violate the lease, a landlord can pursue eviction in civil court. Tenants are allowed to plead their case and resolve the matter in accordance with state laws, though they are not given a court-appointed attorney and failure to appear in court often results in a judgment in favor of the landlord. If the court allows the eviction and tenants are still residing in the home, they may be removed by law enforcement or other officials. Very few eviction filings follow such a linear path, and only a small portion end with forcible removal by law enforcement. Some tenants choose to leave the property on their own, others resolve the violation outside of court, and illegal evictions are never filed in the first place; landlords can and do perpetrate illegal evictions by changing locks, physically removing tenants, and leveraging threats. No matter the scenario, tenants already under financial strain often do not have resources to adequately plead their case in court.

In many cases, the mere filing of an eviction in court goes on a tenant's rental record, whether or not the eviction was completed. These records can be screened by landlords who do not rent to tenants with prior evictions, making future housing more difficult to find. Once a tenant is on this path, possibly due to a single, incomplete filing, it becomes harder and harder to secure stable housing. Social scientists know that this housing insecurity permeates every corner of a family's life: it may decrease health outcomes due to stress and reduced access to medical care, induce job-loss and reduced earnings, and even directly lead to homelessness (Cornec, 2023). In this way, eviction is not merely a result of poverty; it is also a cause.

What (and How) Do We Know About Eviction?

Though millions of Americans face life-altering evictions every year (Garnham, Gershenson, & Desmond, 2022), the federal government does not compile data on evictions. States and localities may individually track evictions, but the data is often difficult if not impossible to locate and inconsistent both between and within states. A large reason for this dearth of data is that it originates in local courts. An official eviction is a legal event that must be filed with a local court system, at which point local clerks become responsible for the records. Some localities track the data in a centralized system, some manually, and some not at all (Vallejo, 2024). And whatever data states do have only represent legal evictions; in 2017, the American Housing Survey estimated that of the 5.3% of forced moves attributed to evictions, 4.5% were informal. This statistic may even underrepresent the rate of informal and illegal evictions in the United States (U.S. Government Accountability Office, 2024).

Despite these obstacles, local and national efforts to aggregate eviction data do exist. Most prominently, The Eviction Lab at Princeton University compiles data from a multitude of sources and employs statistical methods to paint a picture of eviction rates across the country. Through these efforts, social scientists know that eviction rates are growing, even in suburban and rural communities (Hepburn, Grubbs-Donovan, & Hartley, 2024). They also know that evictions are often filed against the same address multiple times, and that a small subset of "serial evicting" landlords are responsible for a large share of eviction filings (Leung, Hepburn, & Desmond, 2020). State and local laws play a large role in defining the eviction pattern across states as well; court fees range widely, and low fees make evictions enticing for landlords (Gomory et al., 2023).

The reason for increasing eviction rates remains debated. While many have historically pointed to gentrification as a compounding factor, emerging research indicates that evictions may actually be concentrated in neighborhoods that are not gentrifying, but are instead perpetually low-income (Hepburn, Louis, & Desmond, 2023). Given that a large bulk of eviction cases are filed by repeat landlords, this makes intuitive sense; landlords are not filing to displace old residents in favor of new ones, but rather to repeatedly remove any tenant, often to their financial gain. Indeed, it is no surprise the individuals and families most impacted by eviction are low-income. Renters facing eviction are disproportionately Black, though there is some speculation that Hispanic renters may experience a high proportion of illegal, undocumented evictions due to immigration concerns in courts. Additionally, children and those living with children face some of the highest risk for eviction—more than double the risk for adults living without children (Graetz et al., 2023). For these reasons, eviction is not merely an economic issue, but one of racial and social justice.

What Does Eviction Look Like in the DMV?

Eviction laws vary widely between cities and states, meaning related data and research is also location-specific. Understanding eviction in a city and its surrounding area requires an understanding of the regulations in play. In 2020, housing sociologists Eva Rosen and Brian McCabe published a report detailing every eviction in Washington D.C. from 2014 to 2018: An average of 32,000 eviction filings per year impacting approximately 18,000 households in the city. Their work mirrors much of Princeton's Eviction Lab; they emphasize the presence of serial eviction filings, the risk of tenants in subsidized housing, and disproportionate impact on marginalized neighborhoods such as Ward 7 and 8. They also note that the vast majority of evictions are for non-payment of rent, and that D.C.'s fee to file an eviction is merely \$15—one of the lowest across the country (McCabe & Rosen, 2020).

Next door, Virginia has historically had some of the highest rates of eviction nation-wide, and though the rates dropped during COVID-19 due to eviction-moratoriums, they have already surpassed pre-pandemic levels in several counties bordering D.C. including Arlington, Alexandria, and Fairfax (Nolan et al., 2024). These numbers are likely due in part to the fact that a large portion of renters are rent-burdened; that is, they spend more than 30% of their income on housing, sometimes more than 50%. Virginia's legal eviction process is also much shorter than many other states—an eviction can be completed from start to finish in fewer than 50 days—giving tenants less time to prevent the official eviction (Housing Opportunities Made Equal of Virginia, 2024). Maryland, too, has very high rates of eviction relative to other states. Similar to D.C., the fee to file an eviction case is very low—sometimes only \$15 (Weingarten, 2023). Evictions may also be filed immediately upon non-payment of rent with no notice to tenants, expediting the process and making it harder for tenants to fight (National Low Income Housing Coalition, 2022).

How Can Machine Learning Help Us Understand Eviction?

Machine learning models have been implemented in several instances to tackle the challenges of eviction. In response to rent-stabilization laws in New York City, Ye et al., 2019 partnered with NYC officials to identify buildings at risk of landlord harassment—a key mechanism leading to illegal eviction in rent-stabilized apartments. The authors develop multiple models including Logistic Regression, Decision Trees, Random Forests, and Gradient Boosting using city data and public records on historical cases, building features, and landlord characteristics. Ultimately, they increase outreach targeting by 59%. More specific to eviction, Tan, 2020 predicts eviction risk at the census block group level in San Francisco utilizing Multiple Linear Regression, Decision Trees, Random Forests, and Long Short-Term Memory (LSTM) networks. The author found that Random Forests best predicted eviction risk, and the results closely mirrored known hot spots for eviction within the city and risk factors, providing insight into targeted intervention.

Tabar et al., 2022a builds on this prior work by introducing their deep learning model MARTIAN, which forecasts the number of tenants at risk of eviction in the future in Dallas, Texas. Their model, based on the American Community Survey and local public data, outperforms several simple machine learning models. Also in Dallas, the same authors make use of satellite imagery to circumvent the lack of eviction data and identify eviction hotspots (Tabar et al., 2022b). These applications of machine learning to eviction may both improve policy outreach efforts, and tackle the lack of reliable court data.

As discussed, data for these projects are local-specific, necessitating models be built specific to their geographic location. To this date, no work has been done on this front in Washington, D.C., though these existing methods may be drawn upon. While this prior work is technically advanced and computationally expensive, I instead employ simple, baseline models in an effort to begin painting a the landscape of eviction within the D.C. Metropolitan Area. Future will be necessary to increase the detail of this landscape, likely requiring more reliable data.

DATA

Dataset Sources and Region Selection

Target data on eviction for this project comes from Princeton University's Eviction Lab. Specifically, I utilize 2018 block-group proprietary data purchased from LexisNexis, as it is the most recent and granular data available. This source includes data on the number of eviction filings, threats, and judgements in block groups across the United States over several years. As with every source of eviction data, there is likely a very large margin of error, a fact The Eviction Lab acknowledges:

“We purchased proprietary individual records data from LexisNexis Risk Solutions (LexisNexis). LexisNexis conducts automated and in-person bulk collection of records from local civil and housing courts in most states. Automated collection typically involves bulk transfer of electronic records from the courts directly to LexisNexis... Due to the sheer volume of eviction cases filed annually and lack of available electronic bulk records collection across courts, it is not realistic to expect that the proprietary data will include the universe of eviction filings.” (Gromis et al., 2022)

For features, I utilize data from the 2018 American Community Survey 5-year Estimates for my model features. These estimates are created by the U.S. Census Bureau by aggregating 5 years of data in order to create statistically robust estimates at a very granular level, such as census block groups. The survey covers a wide range of topics such as demographics, education, income, housing quality, commute times, etc. However, due to the estimated nature of the values, there may be significant margins of error. I begin with a large, semi-random set of variables from the 2018 5-year ACS. Through my process, this set is narrowed, discussed further below.

Using shape files from the 2020 U.S. Census, I select block groups contained within the Washington D.C. Metropolitan Statistical Area. Specifically, I focus on Washington D.C. and counties directly neighboring the district: Arlington, Falls Church, Alexandria, Fairfax, Prince George's, and Montgomery. The final selection comprises 2,548 census block groups across Virginia, Maryland, and the District of Columbia. I then filter ACS data to only the selected block groups, and merge this ACS data with Eviction lab data by block ID. The resulting dataset consists of 1,498 block groups and 283 potential socioeconomic variables. Figure 1 shows the complete selected region and the subset of block groups with eviction data used in models. Notably, eviction data is available primarily for D.C. and Virginia, with over 70% of block groups represented in each county except Falls Church city, which is comprised of less than 10 block groups. Maryland has much less data available, with only 59% of groups in Prince George's and 6% in Montgomery. It is likely this imbalance between states affects the modeling process.

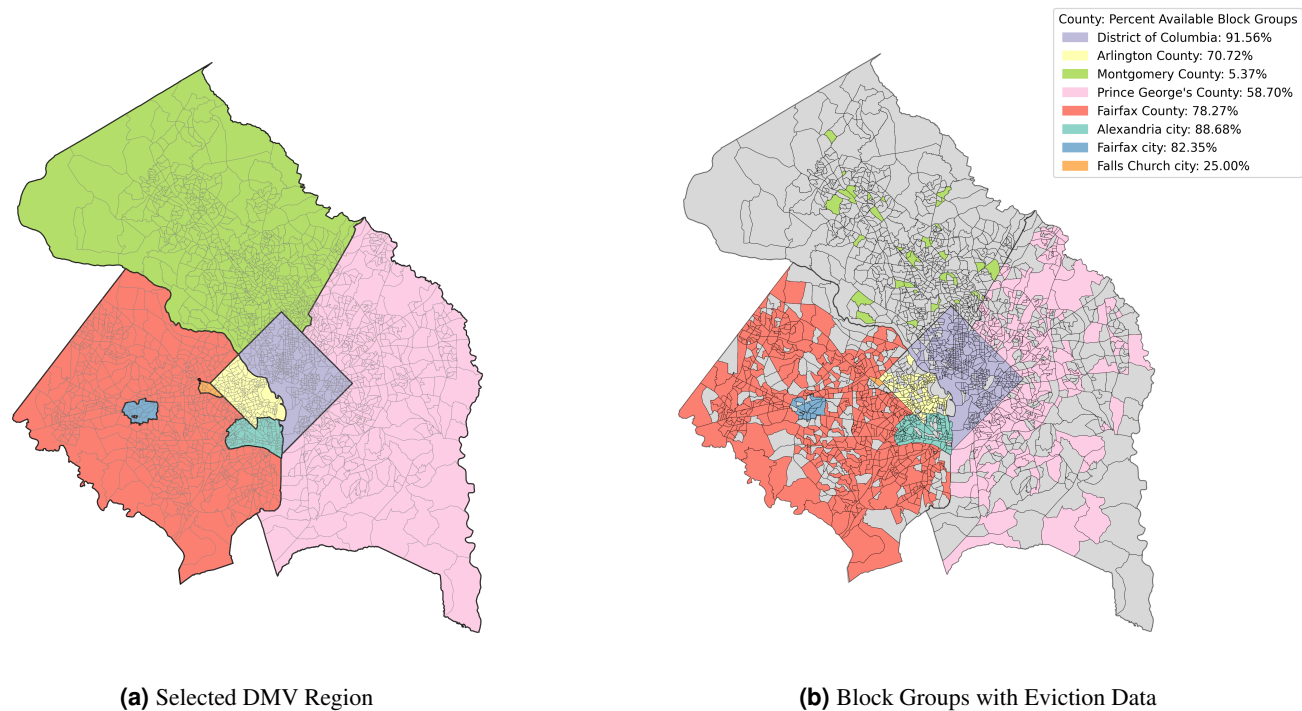


Figure 1. The selected DMV region accounting for 2,548 total census block groups and 1,498 with eviction data. Percentages represent the block groups with eviction data within the given county. Note Falls Church City contains an enclave within Fairfax City.

Preprocessing Feature Variables

I begin with a set of 283 potential features from the 2018 The American Community Survey 5-year estimates. I additionally create indicators for the three states to potentially capture any state-level effects due to differing laws. Of the original 283 features, 86 contained no data and were dropped. 7 additional features were missing more than 50% of their data, and were also dropped as imputation at this scale would likely introduce bias. This leaves 190 features, 25 of which contained anywhere from 0.02% to 30% of their data missing.

To tackle these missing values, I employ multiple imputation by chained equations (MICE). MICE imputes missing values based on the both existing values for an observation and relationships with other observations, creating a flexible approach that accounts for uncertainty within the data (Durbin & Koopman, 2000). Figure 2 shows the distribution of these 25 variables pre- and post-imputation. From here, I standardize all variables in preparation for classification models. Ultimately, I have 190 features across 1,498 census block groups. These cover a wide range of topics from race, education, income, languages, employment, commuting, and housing characteristics.

Preprocessing Target Variables

Though Eviction Lab data covers several measures of eviction, including filings, threats, and judgments, I choose filing rates as my target for several reasons. First, many filings do not result in a judgment, yet still have major impacts on tenants' lives. Overlooking these "incomplete" filings would not capture the true impact and landscape of evictions, or their repercussions. Second, block groups with high rates of filings may identify the location of serial evicting landlords. Overall, I hope to build a model that may aid in *preventing* evictions before they happen, starting with preventing the filing. Notably, there are several ways to support tenants during the eviction process after a filing as well, but for simplicity I chose to focus on this beginning step.

Figure 3 displays the distribution of both total filings and filing rates based on block group population across all 1,498 block groups. Figure 4 additionally displays this information by state. Values in Washington, D.C. consistently outpace Virginia and Maryland even adjusted for population, though this may represent the imbalance in data across states as discussed above. These distributions are extremely skewed to the right, reflecting the presence of block groups with extreme serial evictors, while the vast majority experience only a few filings in the year.

To simplify a preliminary model, I discretize the continuous Filing Rate distribution into a binary target by its 50th percentile. Due to the skewed nature of the variable distribution, this results in two classes balanced by the number of observations (749

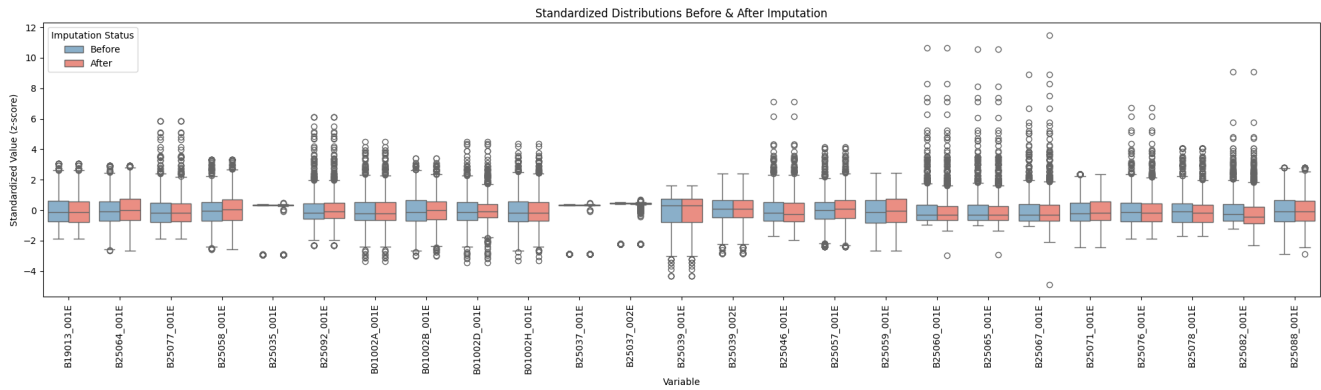


Figure 2. The distribution of 25 variables imputed with MICE before and after imputation. These variables have been standardized.

each), but with disparate ranges where the first class represents eviction rates up to 5.84 and the second any larger rate. I consider these classes "low risk" and "high risk" of eviction. Though this step is certainly an oversimplification of a complex issue, it allows for simple, baseline modeling.

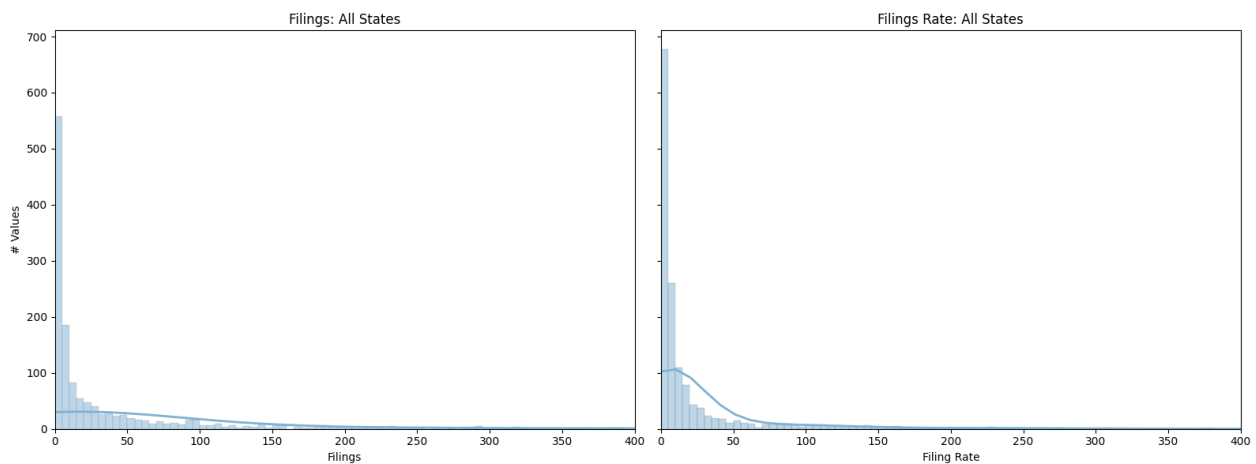


Figure 3. The distribution of filings (left) and filing rates (right) for each block group across all three states.

RESULTS

Logistic Regression

I begin with the simplest of my three binary classification models, logistic regression. This model boasts strong interoperability, making it attractive policy-making where findings must be easily communicated to non-technical audiences. However, this model assumes linear relationships between variables, and may struggle with the multicollinearity inherent in a socioeconomic dataset.

I employ a 5-fold cross-validated grid search on 80% of my data, resulting in a model with strong Elastic Net Regularization $C = 0.01$ and $L1 \text{ Ratio} = .9$. Applied to the test set, this model has an overall accuracy of 73.00%. Table 1 displays performance values for this model with Figure 5 showing the ROC curve and confusion matrix. The logistic regression demonstrates a balanced predictive capability between classes with the number of false positives and false negatives nearly identical.

Random Forest Classifier

I then employ a random forest classifier that I anticipate may better handle nonlinear relationships between features. However, trees are prone to overfitting and must be carefully controlled, as well as being much less interposable relative to logistic regression models. This model is also more computationally expensive.

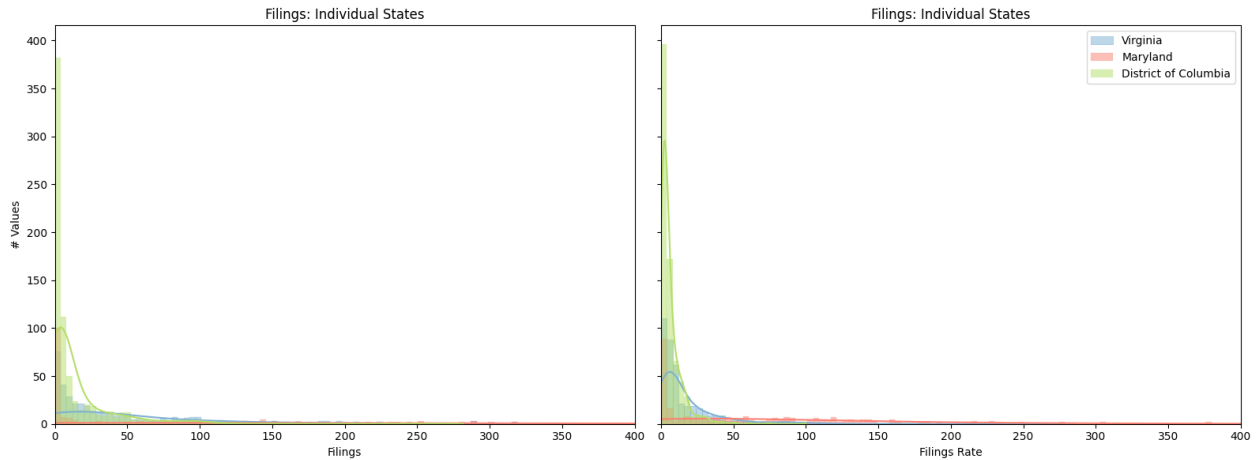


Figure 4. The distribution of filings (left) and filing rates (right) for each block group for individual states comparatively.

Performance Metric	Value
Accuracy	0.7300
Precision	0.7143
Recall	0.7092
F1 Score	0.7117
ROC AUC	0.7957
MSE	0.2700

Table 1. Performance metrics for Logistic Regression

I similarly utilize a 5-fold cross-validated grid search on my training data. The resulting model is constrained by a minimum number of samples for each leaf (2), but has no max depth. Even so, there is little evidence of overfitting. This model has a slightly better overall accuracy of 77.33%. Table 2 displays performance and Figure 6 the ROC curve and confusion matrix for this random forest, showing the strong precision but struggling recall; the increased accuracy is paired with an increased false-negative rate.

Evaluation Metric	Value
Accuracy	0.7733
Precision	0.8067
Recall	0.6809
F1 Score	0.7385
ROC AUC	0.8243
MSE	0.2267

Table 2. Performance metrics for Random Forest Classifier (RFC)

Support Vector Classifier

Last, I apply a Support Vector Classifier, a model that is particularly useful for high dimensional data. Similar to random forests, this model can be difficult to interoperate and is the most computationally expensive of the three.

A 5-fold cross-validated grid search results in a non-linear RBF kernel and moderate regularization at $C = 1$. This adaptive set up indicates non-linear decision boundaries, resulting in an overall accuracy of 74.00%. Table 1 displays performance values and Figure 7 demonstrates the ROC curve and confusion matrix for the SVC model. Similar to the random forest, high precision is paired with low recall.

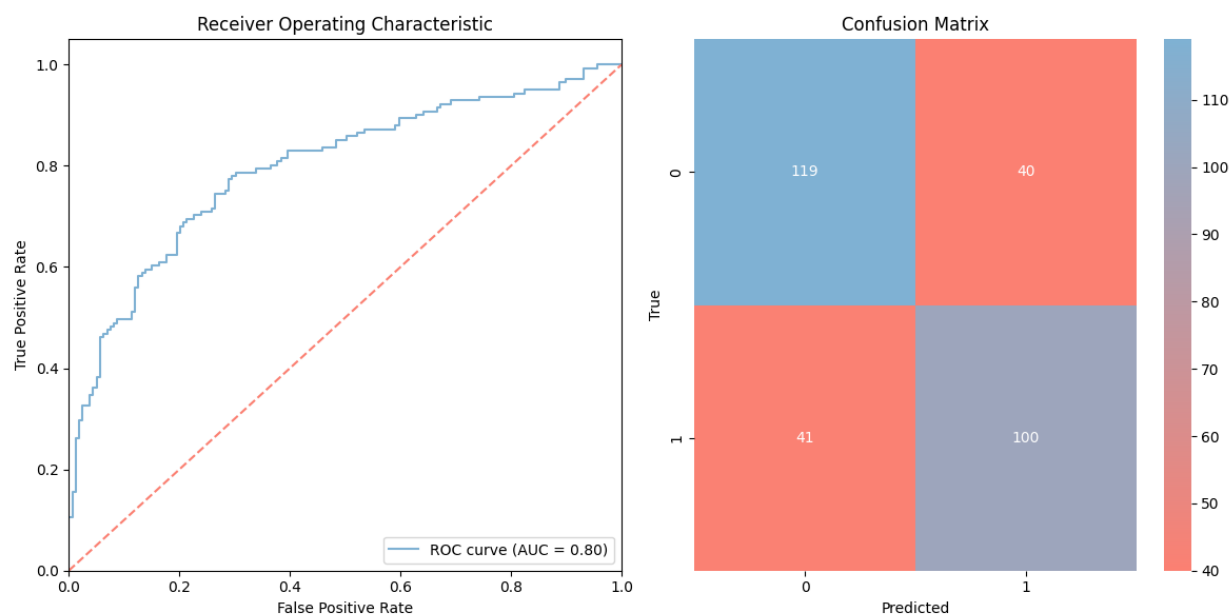


Figure 5. ROC curve (left) and confusion matrix (right) for the logistic regression model

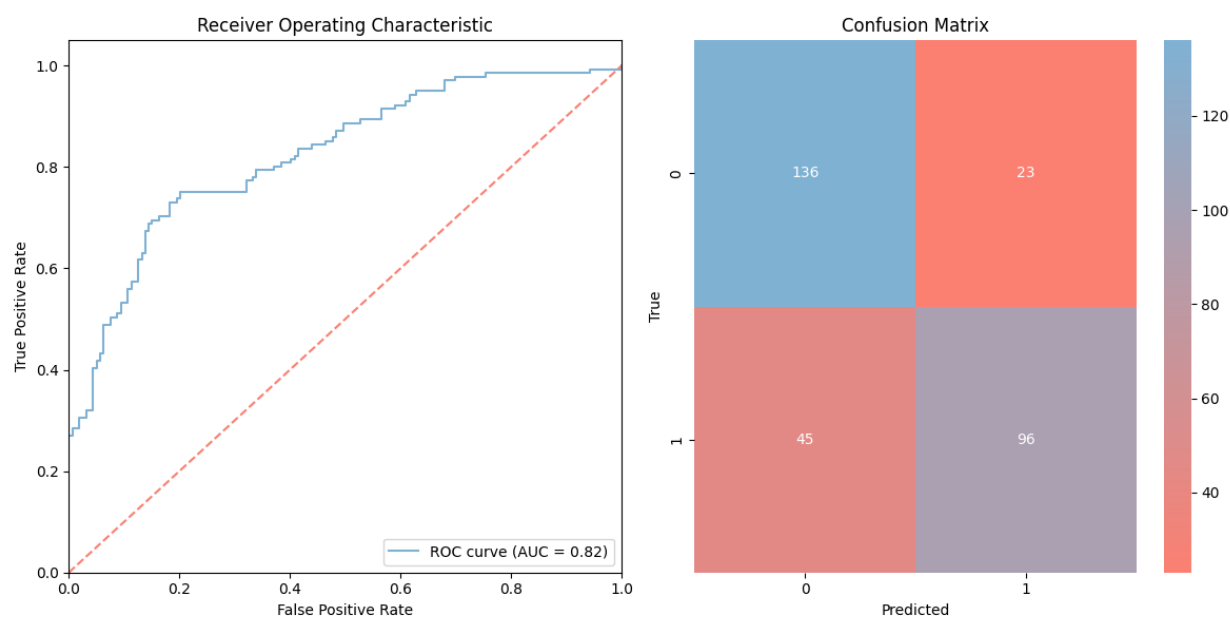


Figure 6. ROC curve (left) and confusion matrix (right) for the random forest model

DISCUSSION

Model Performance

Table 4 displays performance evaluations and confusion matrix values for each model in comparison. The random forest model is the strongest of the three overall, approaching an acceptable 80% accuracy. However, with a recall of only 68%, the model does not identify a large portion of true positives. In this context, a poor recall means block groups with a "high risk" of eviction are not identified, problematic for implementation and targeted assistance. Arguably, the consequences of a false negative—missing a high risk group—far outweigh the consequences of a false-positive—identifying a low risk group as high risk. Particularly when considering intervention efforts, I argue that supporting low risk communities is preferable to not supporting high risk ones, even at a potential monetary loss.

Metric	Value
Accuracy	0.7400
Precision	0.7561
Recall	0.6596
F1 Score	0.7045
ROC AUC	0.8059
MSE	0.2600

Table 3. Performance metrics for Support Vector Classifier (SVC)

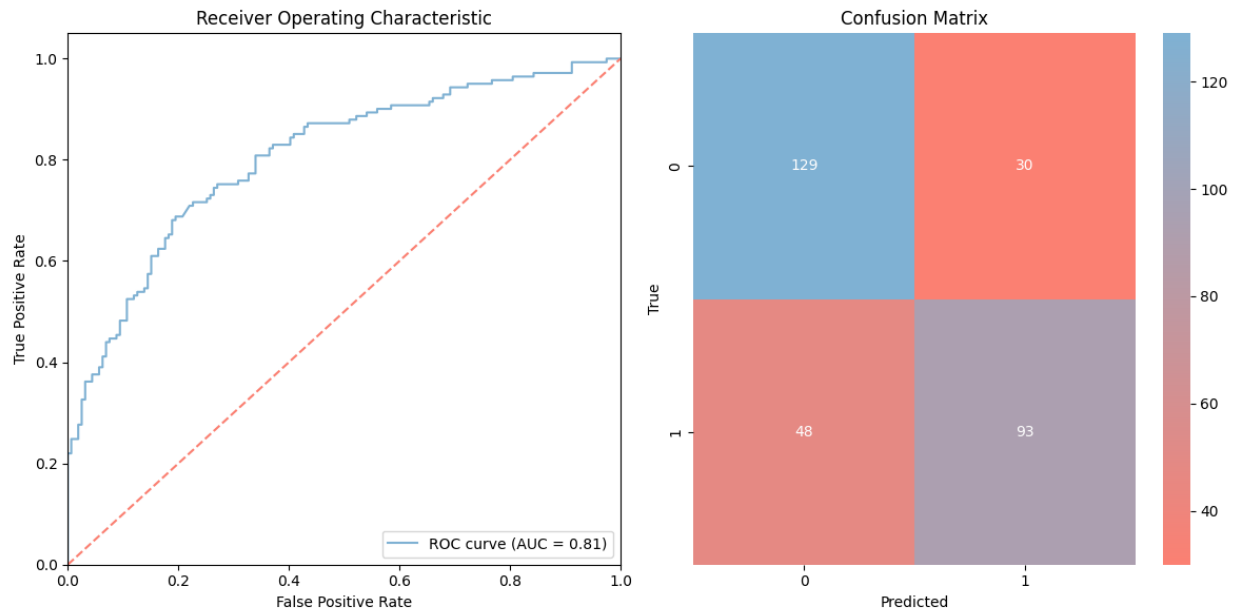


Figure 7. Shows ROC curve (left) and confusion matrix (right) for the support vector model

Under this assumption, I turn to the logistic regression model. Though this model has the lowest accuracy of the three, it also has the highest recall, meaning it is the least likely to overlook high risk block groups. It also identifies the most true positives. If a model such as these is intended to help policy makers implement targeted interventions or identify areas at high risk of eviction, the logistic regression model may in fact be the right choice despite its lower accuracy.

Performance Metric	Logistic Regression	SVC	Random Forest
Accuracy	0.7300	0.7400	0.7733
Precision	0.7143	0.7561	0.8067
Recall	0.7092	0.6596	0.6809
F1 Score	0.7117	0.7045	0.7385
ROC AUC	0.7957	0.8059	0.8243
MSE	0.2700	0.2600	0.2267
TN (True Negatives)	119	129	136
FP (False Positives)	40	30	23
FN (False Negatives)	41	48	45
TP (True Positives)	100	93	96

Table 4. Comparison of Logistic Regression, SVC, and Random Forest Classifier across evaluation metrics and confusion matrix components.

Realistically, none of these models perform well enough to support deployment in their current state. Accuracies between 70-80% indicate there is certainly signal between my features and targets, as well as room for improvement. Such improvement

may take two paths: first, one could attempt to improve accuracy of the logistic regression without sacrificing recall, and second, one could attempt to improve the recall of either the Support Vector or Random Forest models without sacrificing accuracy. Given the strong correlations between feature variables and non-linearity demonstrated by the more computationally expensive models, the second option is likely more viable. A third option is to test entirely new models such as XGBoost or a small neural network, which may be better equipped to handle expansive and geography-specific socioeconomic data. More work is needed on any of these fronts to improve modeling and assert its usefulness.

Key Features

Table 5 contains 28 key predictors derived from the logistic regression model. While the model's accuracy certainly needs improving, evaluating these predictors may still provide insight on evictions in the DMV. Notably, the most significant predictor of eviction filings is being located in Washington, D.C. This may reflect demographic differences between D.C. and its bordering states, with a higher Black and Hispanic residents as well as general urbanicity trends. The second most significant predictor is the presence of presumably single-mother households. Literature on the topic indicates that single-mothers often find themselves in the lowest income groups, making this outcome unsurprising, if distressing.

Additionally, several known demographic indicators appear in these key variables, including race, education, and financial status. Block groups with high aggregate and median rents as well as a large number of households paying a significant portion of their income on rent are more likely to be "high risk". Potentially counter to this trend is the number of renters with income below 10,000 actually decreasing risk, though this group may be receiving forms of public support. Also noteworthy is the impact of race; Black populations drive up eviction risk as well as groups that speak Spanish as a first language, potentially a proxy Hispanic populations. Asian populations, on the other hand, have the strongest negative affect on eviction risk, followed closely by white populations. There is also a clear trend in education, where higher attainment correlates with decreased risk and lower attainment increased risk. Also noteworthy and unsurprising is the influence of households receiving some form of public assistance.

Several surprising features also emerge in this analysis, though they may be theoretically explained. First, several features regarding "owner-occupied" units are significant, though in countering directions. 1 person owner occupied units and attached 1 unit housing both decrease eviction risk, while 1-unit detached housing increases the risk somewhat drastically. These features may indicate a few things about the block group. First, the presence of owner-occupied households will intrinsically decrease the number of renters in a group, decreasing the number of potential evictions by nature. Our target rate of eviction filings accounts for the population in a location, but not necessarily the number of renters. On the flip side, 1-unit detached households are often more expensive than apartment units and condos, potentially increasing the financial burden on renters in groups where this form of housing dominates.

Similarly, the year housing was built appears several times in our variables, possibly also acting as a proxy for neighborhood cost. In groups where the median year of building is more recent, eviction risk increases slightly, while older housing units (1980-1999) decrease risk slightly. In neighborhoods like Georgetown within Washington, older houses are often more expensive, though the market also sees an influx of expensive new apartments in neighborhoods like NoMa as well, so trends may be hard to decipher. A recent influx of newly built "affordable" housing may also contribute to the increase of risk for new buildings. A closer analysis on these trends is needed to say for certain.

Overall, the story told by this feature analysis is unsurprising; single mothers, Black and Hispanic communities, neighborhoods with poor educational attainment, and low-income or assisted groups all appear to have a higher risk of eviction filings. These facts reflect well known trends in literature on poverty, class, and urbanicity. Potentially, other features such as those reflecting owner-occupied units and building year are also proxies for characteristics like housing cost and neighborhood status. It is important to note that these features are *not* causal, but correlational, and more analysis is needed to substantiate directional trends and their magnitudes.

Block Groups At Risk

Figure 8 displays the predicted risks for all block groups across each model. Approximately the same number of block groups are classified as high eviction by the logistic regression (45.66%) and support vector (44.33%) models. The random forest model predicts slightly more block groups at 50.13%, primarily represented in Washington, D.C.

Unfortunately, block groups are a census derived geographical aggregation for the purpose of statistical analysis, and do not necessarily represent communities or neighborhoods, making it difficult to draw conclusions from this form of analysis. Primarily, it is clear that all three models weight Virginia heavily, somewhat surprising given the value assigned to D.C. in the logistic regression model. This may be a result of the imbalance between states in the initial dataset, or possibly a reference to lax laws in VA, though this option seems less likely.

Feature Name	Coefficient	Odds Ratio	Description
state_11	0.4913	1.6344	Washington, D.C.
B11001.006E	0.3568	1.4287	Family households with a female householder and no husband present
B25032.003E	0.2601	1.2971	Owner-occupied 1-unit detached housing (i.e., standalone houses)
B25070.008E	0.1840	1.2020	Renter-occupied housing units paying 35–39.9% of their income on rent
B25065.001E	0.1697	1.1849	Aggregate gross rent paid by renter-occupied housing units paying "cash rent"
B16004.006E	0.1673	1.1822	Population aged 5–17 that speaks Spanish at home and speaks English "well"
B16004.007E	0.1607	1.1744	Population aged 5–17 that speaks Spanish at home and speaks English "not well"
B19057.002E	0.1092	1.1153	Households receiving public assistance income in the past 12 months
B15003.017E	0.0852	1.0889	Population aged 25+ with a high school diploma (excluding GED)
B25071.001E	0.0824	1.0859	Median gross rent as a percentage of household income for renter-occupied units paying "cash rent"
B16004.003E	0.0698	1.0723	Population aged 5–17 that speaks only English
B02001.003E	0.0696	1.0721	Black or African American population
B15002.009E	0.0649	1.0670	Men 15+ whose highest education is 11th grade
B25037.002E	0.0637	1.0658	Median year built of owner-occupied housing units
B08303.004E	-0.0685	0.9338	Workers 16+ who commute 10–14 minutes (non-work-from-home)
B25074.008E	-0.0800	0.9231	Renters with <\$10,000 income paying 40–49.9% on rent
B25034.005E	-0.0869	0.9168	Housing units built 1990–1999
B25034.006E	-0.0924	0.9117	Housing units built 1980–1989
B25032.004E	-0.1004	0.9045	Owner-occupied 1-unit attached housing
B08303.003E	-0.1146	0.8917	Workers 16+ who commute 5–9 minutes (non-work-from-home)
B15003.022E	-0.1175	0.8891	Population 25+ with a bachelor's degree as highest education
B16004.005E	-0.1201	0.8868	Population aged 5–17 that speaks Spanish at home and English "very well"
B25040.002E	-0.1604	0.8518	Occupied housing units heated with gas
B15003.025E	-0.1706	0.8431	Population 25+ with a doctorate as highest education
B03002.003E	-0.2007	0.8182	White (not Hispanic or Latino) population
B25009.003E	-0.2121	0.8089	Owner-occupied units with 1-person households
B25059.001E	-0.2363	0.7896	75th percentile of contract rent for units paying "cash rent"
B02001.005E	-0.2413	0.7856	Asian population

Table 5. Top Feature Coefficients and Odds Ratios from the Logistic Regression Model. Note that the phrase "cash rent" implies a fixed, monetary rent payment that is not free or public assisted. It does not mean a physical cash payment.

Limitations

This project has several important limitations. First, data on eviction is overwhelmingly flawed and arguably unreliable. Utilizing data from a proprietary source such as Lexis means the collection process is unknown. Additionally, Eviction Lab data does not exist for all of my selected region, resulting in imbalanced data across states and missing data that is potentially important. For a stronger analysis, data on evictions and filings would need to be more accurate, granular and widespread

Second, ACS 5-year data is known to be imprecise in many instances, particularly on a small scale such as block groups. These estimates are aggregated across 5 years and still have varying margins of error that influence modeling. Additionally, I preformed a semi-random selection of feature variables from ACS data and did not account for margin of error on estimates. Future work should more carefully consider feature variables and their errors.

Third, all of the data utilized is dated 2018, making the analysis slightly out of date. For such fast moving markets such as rental and housing, this delay can drastically influence outcomes. More up-to-date data is necessary for policy-relevant decisions. 2018 also pre-dates the COVID-19 pandemic, an influential event on all topics considered here.

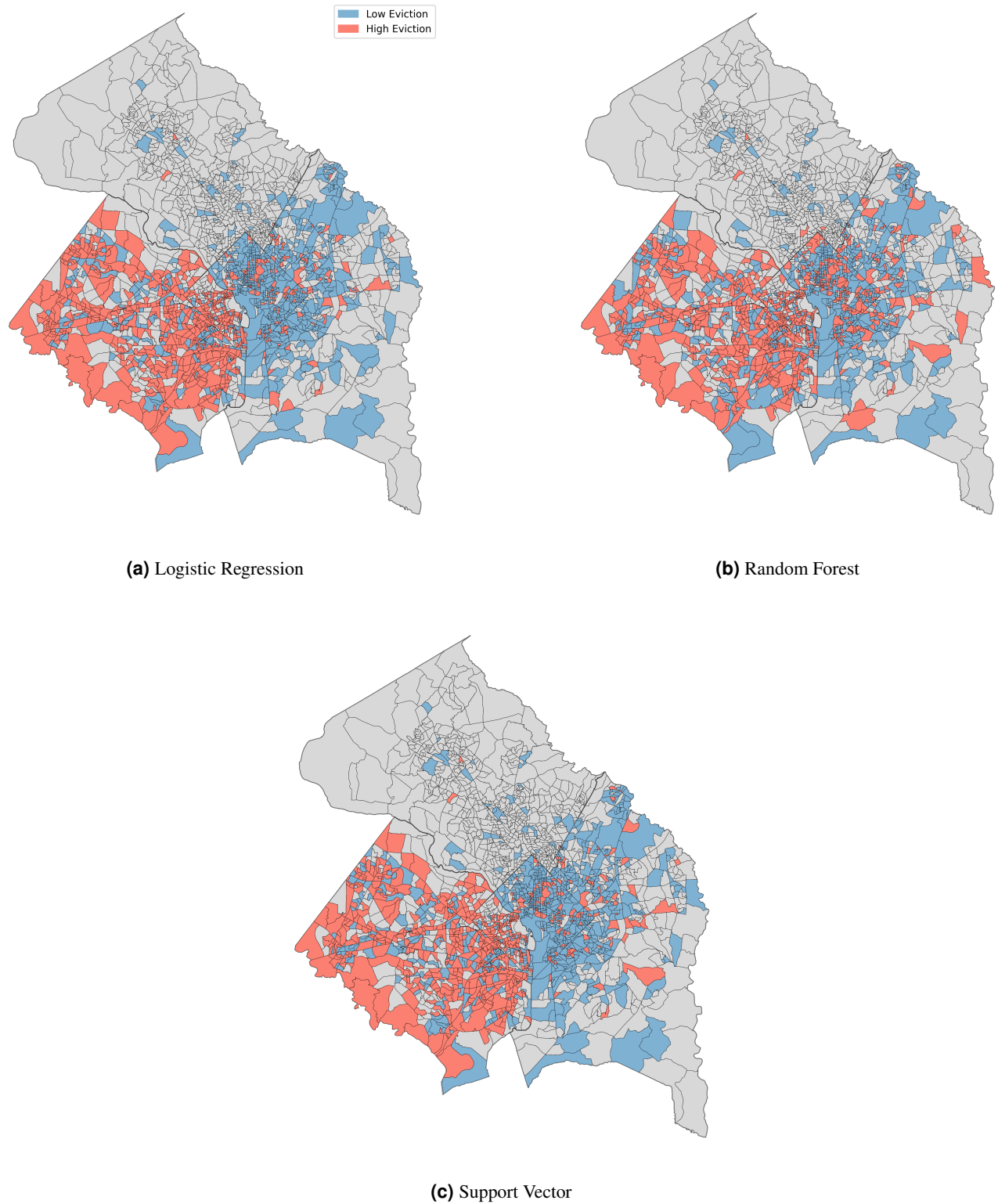


Figure 8. Predicted eviction values for each classification model.

Any conclusions from this project are limited by model accuracy, which is strong but certainly needs improving. At this juncture, I can not recommend any decisions or offer future predictions for this reason. As discussed, more model-tuning and possibly different models are necessary to improve accuracy and outcomes.

CONCLUSION

In this project, several machine learning models were employed to predict eviction risk across a subset of the Washington D.C. Statistical Area. Utilizing ACS estimates and Eviction Lab Data, a binary target was created measuring eviction filings within block groups. Three classification models (logistic regression, random forest classification, and support vector classification) were employed with accuracies ranging from 73.00% to 77.33%. While the random forest model preformed strongest across most metrics, it struggled with recall, a key metric for reducing false negatives and targeting communities at risk of eviction. On this front, the logistic regression model outperformed the two remaining

A preliminary feature analysis identified several key predictors of eviction risk in alignment with existing literature on the topic. Socioeconomic predictors such as income, race, and education were all influential, with single-mother households increasing the likelihood of a "high risk" classification by 40%. Several other factors such as owner-occupied status and building year also emerged as significant features, possibly as proxies for housing quality and neighborhood cost. These results confirm prior understanding of eviction as an issue and driver of poverty and social justice.

However, limitations in data sources and availability drive caution regarding the magnitude and accuracy of these results, as well as clear bias in predictions across different locations. More reliable data is needed for future analysis. As eviction is a location-specific issue, work should ideally be done within cities and states to capture geography-specific trends in laws and enforcement. However, machine learning applications necessitate large data sets, and as such granular data within cities and states that is often unavailable.

Nevertheless, this project demonstrate that machine learning can be a powerful tool for understanding eviction patterns, providing valuable insights for policymakers and community stakeholders. These predictions have the potential to inform targeted interventions aimed at reducing eviction rates and addressing housing instability. Future work could explore the integration of additional data sources and the refinement of model techniques to further enhance the accuracy and generalization of these predictions. Forecasting models may prove particularly useful in targeting prevention efforts. An eviction is a traumatic, long-lasting event for a family that influences many aspects of their life and necessitates institutional attention. Projects such as this one may help achieve this goal.

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